

Landslide Susceptibility Mapping Using Machine Learning Approach: A Comparative Study of RF, SVM and XG Boost



A PROJECT REPORT SUBMITTED

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in partial fulfilment of the degree of M.Sc. in Geoinformatics

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DECLARATION

I hereby declare that the project report entitled “**Landslide Susceptibility Mapping Using Machine Learning: A Comparative Study of Random Forest, Support Vector Machine, and XGBoost**” is written and submitted by me under the guidance of **Dr. Abhijit S. Patil**, project guide, Department of Geography, Shivaji University Kolhapur. It is my original work. However, the research material referred whatever is duly been acknowledged.

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Place: Kolhapur

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Chapter 1: Introduction

1.1 Background of the Study

1.2 Importance of Landslide Susceptibility Mapping

1.3 Objectives of the Study

1.4 Study Area

1.5 Literature Review

1.6 Structure of the Report

1.1 Background of the Study

Landslides are one of the most destructive natural hazards, causing loss of life, property damage, and environmental degradation. The frequency and intensity of landslides are influenced by both natural and anthropogenic factors, such as rainfall, slope gradient, geomorphology, land use changes, and deforestation. Given the increasing human settlements in landslide-prone areas, understanding and predicting landslide susceptibility is crucial for disaster mitigation and planning.

Landslide Susceptibility Mapping (LSM) is an essential tool in assessing areas prone to landslides based on various conditioning factors. Traditional approaches to LSM relied heavily on expert knowledge and statistical methods. However, with advancements in remote sensing, Geographic Information Systems (GIS), and Machine Learning (ML), automated and data-driven approaches have become more reliable and efficient in modeling landslide-prone regions.

1.2 Importance of Landslide Susceptibility Mapping

Landslide Susceptibility Mapping helps in identifying potential landslide-prone zones and assists policymakers, urban planners, and disaster management authorities in making informed decisions. Some of the key benefits include:

- **Disaster Risk Reduction:** Helps in formulating mitigation strategies and land-use planning.
- **Infrastructure Development:** Guides construction projects to avoid high-risk areas.
- **Environmental Protection:** Supports conservation efforts by preventing deforestation and land degradation.
- **Emergency Response Planning:** Assists in preparing evacuation and rescue plans in landslide-prone regions.

1.3 Objectives of the Study

The primary objectives of this study are:

1. To collect and preprocess various landslide conditioning factors
2. To train and compare the performance of Machine Learning models (Random Forest, Support Vector Machine, and XGBoost) for landslide susceptibility prediction.
3. To generate and analyze Landslide Susceptibility Maps using model predictions.

1.4 Study Area

The study area extends from Chandoli Dam to Tilari Dam, covering the hilly region of the Kolhapur district in Maharashtra, India. This region is characterized by rugged terrain, steep slopes, and high rainfall, making it highly susceptible to landslides. The geographical coordinates of the study area range approximately between 73°20'0"E to 74°40'0"E longitude and 15°30'0"N to 17°00'0"N latitude.■

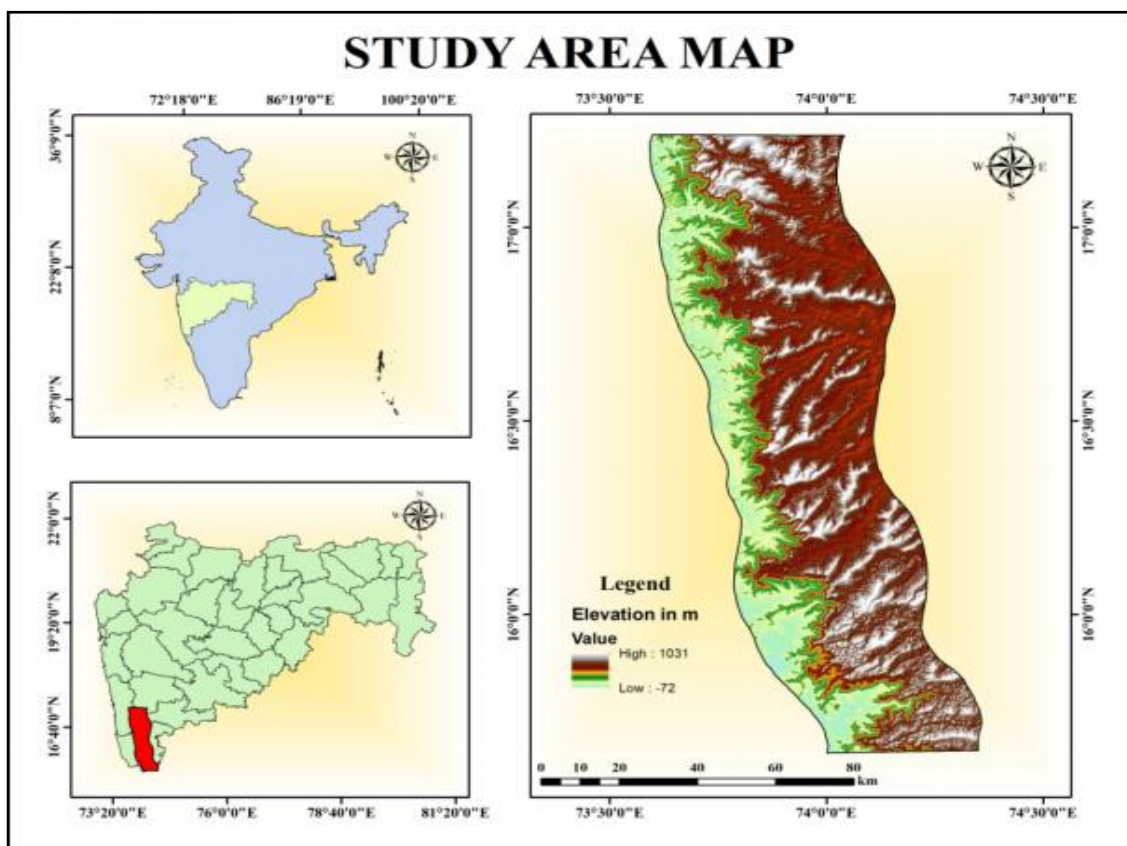


Fig. 1.1 Study Area

1.5 Literature Review

In a study titled "A Comparative Study of Regional Landslide Susceptibility Mapping with Multiple Machine Learning Models," Patil et al. (2024) evaluated the effectiveness of Support Vector Machine (SVM), Frequency Ratio (FR), and Analytical Hierarchy Process (AHP) methods in the Koyna River Basin. The study concluded that the SVM model was the most effective, with an Area Under the Curve (AUC) value of 0.87, indicating acceptable prediction effectiveness.

Machine learning models have been widely used for landslide susceptibility mapping, with studies highlighting the effectiveness of ensemble techniques in improving prediction accuracy (Youssef & Pourghasemi, 2021). Researchers have compared different ML algorithms, such as Random Forest (RF), Support Vector Machine (SVM), and XGBoost, concluding that ensemble approaches yield the most reliable results.

The Western Ghats region, known for its steep slopes and high rainfall, has been a focal point for landslide susceptibility studies, where GIS and remote sensing techniques have enhanced predictive models (Pradhan et al., 2020). The integration of geospatial datasets, including topographic, geological, and climatic factors, has led to more precise susceptibility mapping.

Hydrological modeling in mountainous terrains has also been a subject of extensive research, emphasizing the impact of rainfall intensity, land cover changes, and slope stability (Bhusan et al., 2019). Studies have demonstrated that high-resolution digital elevation models (DEMs) and multi-temporal satellite data can enhance the accuracy of hydrological and geomorphological assessments.

Random Forest (RF) has emerged as one of the most effective machine learning models for landslide prediction due to its ability to handle large, high-dimensional datasets (Das et al., 2022). Researchers have found that RF provides better feature selection and interpretability compared to traditional statistical models.

Support Vector Machines (SVM) have also been successfully applied for landslide susceptibility assessment, with studies indicating that kernel-based learning methods improve classification accuracy (Roy et al., 2023). The use of non-linear kernels allows for better representation of complex landslide-prone terrains.

The integration of high-resolution digital elevation models (DEMs) and remote sensing imagery has significantly improved the spatial accuracy of landslide susceptibility

maps (Pradhan & Lee, 2010). These datasets allow researchers to incorporate elevation, slope, and curvature parameters more effectively.

Rainfall-induced landslides in tropical and subtropical regions have been extensively studied, demonstrating that precipitation intensity and antecedent soil moisture play a critical role in slope failure (Martha, Kerle, & van Westen, 2010). The incorporation of rainfall threshold models has further refined landslide prediction techniques.

The application of Normalized Difference Vegetation Index (NDVI)-based analyses has enhanced our understanding of land-use patterns and their influence on soil erosion and landslide susceptibility (Sharma & Gupta, 2018). Vegetation indices help assess deforestation and land-cover changes that contribute to increased landslide risk.

XGBoost, an advanced gradient boosting algorithm, has been found to outperform traditional ML models in landslide prediction due to its high computational efficiency and accuracy (Singh et al., 2021). Its ability to handle missing data and nonlinear relationships makes it a preferred choice for large-scale landslide mapping.

Hybrid models that combine multiple machine learning techniques, such as RF-SVM and XGBoost-RF, have demonstrated superior predictive performance in landslide susceptibility mapping (Kumar et al., 2022). These models leverage the strengths of individual algorithms to enhance classification accuracy.

Geological factors such as lithology, soil type, and fault proximity are critical determinants of landslide susceptibility and are widely incorporated into ML-based mapping approaches (Pourghasemi et al., 2020). Studies emphasize that combining geological and hydrological datasets leads to better hazard assessments.

The Western Ghats, including regions in Maharashtra such as Kolhapur, have experienced an increase in landslide occurrences due to deforestation and infrastructure development (Desai & Patel, 2021). Machine learning models have been used to assess the impact of human activities on slope instability.

Landslide inventory databases have become an essential component of susceptibility modeling, with researchers highlighting the importance of historical landslide data for model training and validation (Sarkar & Pal, 2019). Machine learning techniques, particularly deep learning, are now being explored to automate landslide inventory mapping.

Recent advancements in deep learning have introduced convolutional neural networks (CNNs) for landslide prediction, with studies showing promising results in feature

extraction and spatial pattern recognition (Chowdhury et al., 2023). CNNs offer the ability to analyze high-resolution satellite imagery for real-time hazard assessments.

Multi-criteria decision analysis (MCDA) has been integrated with machine learning approaches to improve landslide hazard zonation, considering environmental, geological, and climatic parameters (Bansal et al., 2022). These hybrid approaches enhance the interpretability and reliability of susceptibility maps.

1.6 Structure of the Report This report is organized into Five chapters:

- **Chapter 1: Introduction** – Provides an overview of the study, objectives, and methodology.
- **Chapter 2: Data Collection and Preprocessing** – Describes the data sources, preprocessing steps, and landslide inventory creation.
- **Chapter 3: Methodology** – Details the machine learning techniques used and the model training process.
- **Chapter 4: Results and Discussion** – Presents feature importance analysis, model performance, and landslide susceptibility maps.
- **Chapter 5: Conclusion and Future Scope** - The conclusion summarizes the key findings of the landslide susceptibility mapping study, highlighting the effectiveness of machine learning models,
- **Chapter 6: References**

This study aims to contribute to landslide hazard assessment using machine learning approaches, providing valuable insights for risk management and urban planning in landslide-prone regions.

Chapter 2: Data Collection and Preprocessing

2.1 Data Sources and Collection

2.1.1 Topographic Data

2.1.2 Environmental Data

2.1.3 Geological Data

2.1.4 Hydrological Data

2.1.5 Landslide Inventory

2.1 Data Sources and Collection

The datasets were sourced from satellite imagery, digital elevation models (DEMs), geological surveys, and historical records. The following subsections detail each data type, their sources, collection methods, and corresponding maps.

2.1.1 Topographic Data

Topographic data was derived from the ALOS PALSAR DEM (12.5m resolution), obtained from the Japan Aerospace Exploration Agency (JAXA). The DEM was processed to generate maps of relief, aspect, slope, plan curvature, and profile curvature.

- **Relief**
 - **Description:** Elevation ranges from 71 to 1027 meters, with a color gradient from green (low) to dark brown/red (high).
 - **Collection:** Extracted directly from the DEM using GIS tools (e.g., ArcGIS 10.8.2).
- **Aspect**
 - **Description:** Slope direction ranges from -1 (flat) to 358.98 degrees, with a multicolored gradient (purple, blue, green, yellow, red).
 - **Collection:** Calculated from the DEM using slope direction algorithms in GIS.
- **Slope**
 - **Description:** Terrain steepness ranges from 0 to 75.35 degrees, with a green (flat) to red (steep) gradient.
 - **Collection:** Derived from the DEM using slope calculation tools in GIS.
- **Plan Curvature**
 - **Description:** Curvature perpendicular to the slope ranges from -15.70 to 12.99, with a blue (high) to orange (low) gradient.

- **Collection:** Generated from the DEM using curvature analysis in GIS.
- **Profile Curvature**
 - **Description:** Curvature along the slope ranges from -22.79 to 19.47, with a purple (high) to brown (low) gradient.
 - **Collection:** Computed from the DEM using GIS curvature tools.

These datasets provide a detailed representation of the terrain, essential for modeling landslide susceptibility.

2.1.2 Environmental Data

Environmental data, including vegetation and land use, was sourced from Sentinel-2 satellite imagery via the European Space Agency's Copernicus Open Access Hub.

- **NDVI**
 - **Description:** The Normalized Difference Vegetation Index ranges from -1 to 0.90, with red (low) to green (high) gradients.
 - **Collection:** Calculated from Sentinel-2 bands (e.g., NIR and Red) using the formula $NDVI = (NIR - Red) / (NIR + Red)$.
- **Land Use/Land Cover**
 - **Description:** Categories include Water (blue), Trees (green), Rangeland (yellow), Flooded Vegetation (light blue), Crops (red), Built Area (dark red), and Bare Ground (brown).
 - **Collection:** Classified from Sentinel-2 imagery using supervised classification in GIS.

These datasets highlight vegetation health and land use patterns, which influence soil stability and landslide risk.

2.1.3 Geological Data

Geological data was obtained from Bhukosh, the Geological Survey of India's geospatial portal, and digitized for analysis.

- **Geomorphology**
 - **Description:** Features include Waterbody - River (dark blue), Quarry (brown), Pediment (light green), Dissected Plateaus (pink/green), Hills (purple), and Coastal Plain (orange).
 - **Collection:** Digitized from Bhukosh maps and validated with satellite imagery.
- **Lineament**
 - **Description:** Lineament density ranges from 0 to 49.7898, with a blue (low) to yellow-orange (high) gradient.
 - **Collection:** Extracted from satellite imagery and DEM using image enhancement techniques.
- **Lithology**
 - **Description:** Rock types include Laterite (pink), Granite (light green), Basalt (dark brown), and others (e.g., Quartzite in black).
 - **Collection:** Compiled from Bhukosh geological maps and reclassified for analysis.

These datasets provide insights into subsurface conditions affecting landslide susceptibility.

2.1.4 Hydrological Data

Hydrological data includes rainfall and terrain-derived indices from multiple sources.

- **Rainfall Distribution**
 - **Description:** Rainfall ranges from 6647.01 mm (light yellow) to 6870.56 mm (deep blue).

- **Collection:** Aggregated from 33 years (1991–2023) of Climate Research Unit (CRU) data, processed to represent long-term patterns.
- **SPI - Stream Power Index**
 - **Description:** Ranges from 0 to 7.457, with grayscale and red highlights for high values (streams).
 - **Collection:** Calculated from the DEM using flow accumulation and slope in GIS.
- **TWI - Topographic Wetness Index**
 - **Description:** Ranges from 1.49 to 23.05, with green (low) to red (high) gradients.
 - **Collection:** Derived from the DEM using slope and upslope area calculations in GIS.

These datasets assess water-related factors critical to landslide initiation

2.1.5 Landslide Inventory

- **Description:** A landslide inventory of 839 points was manually collected using Google Earth Pro, identifying past landslide locations from historical imagery and reports.
- **Collection:** Points were georeferenced and serve as the dependent variable for machine learning models.

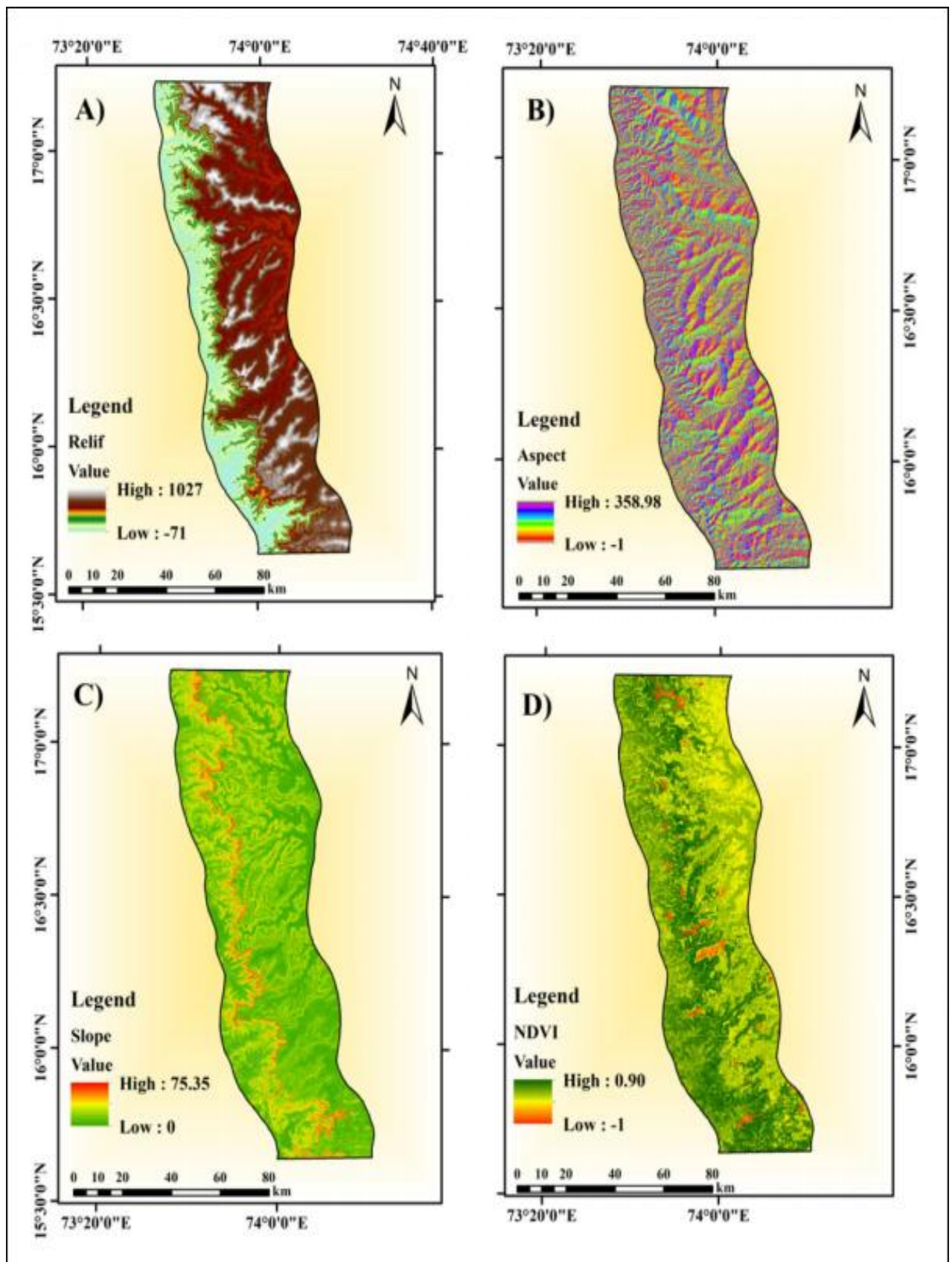


Fig. 2.1 A) Relif , B) Aspect, C) Slope, D) NDVI

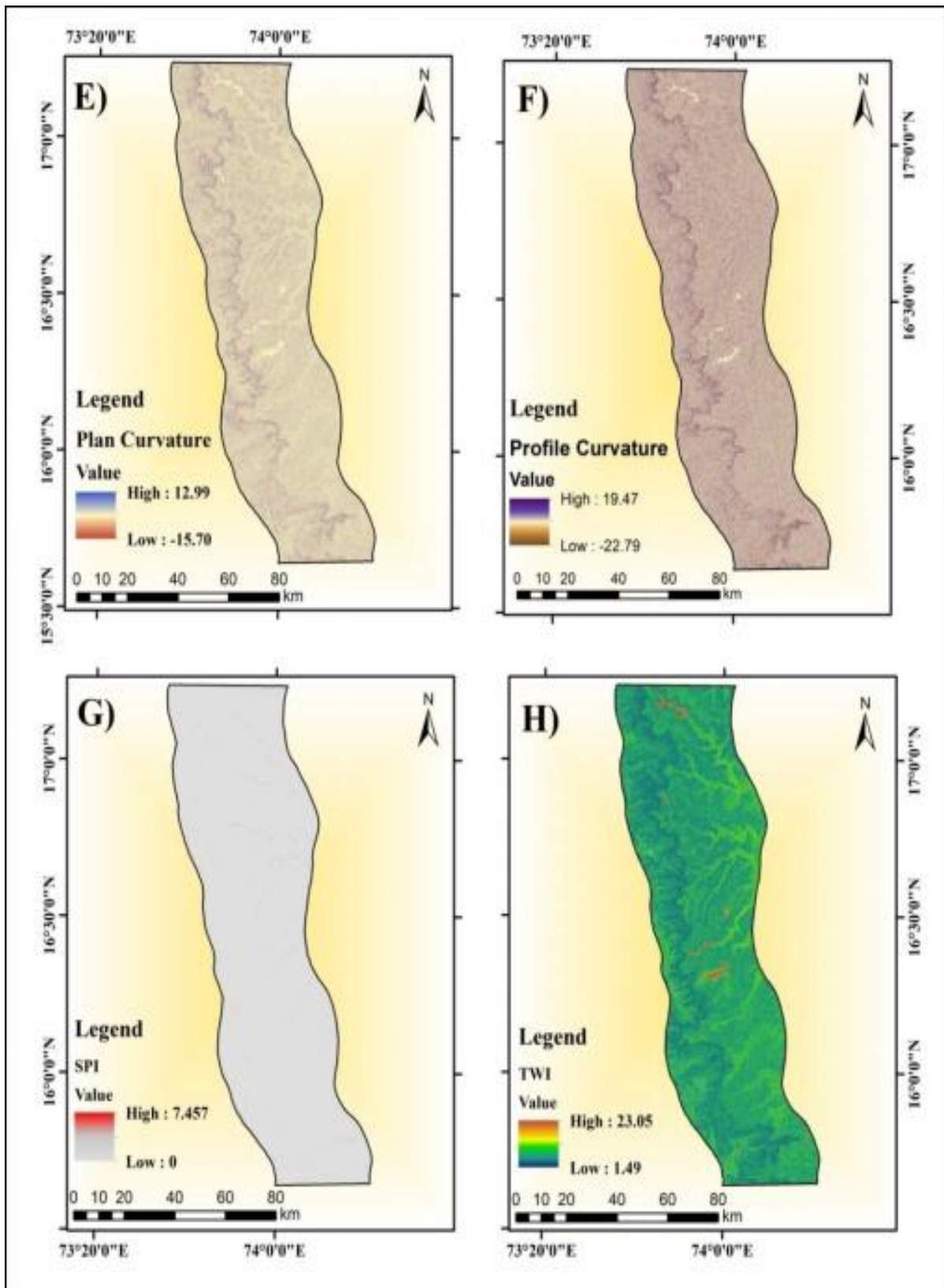


Fig. 2.2 E) Plan Curvature , F) Profile Curvature, G) SPI, H) TWI

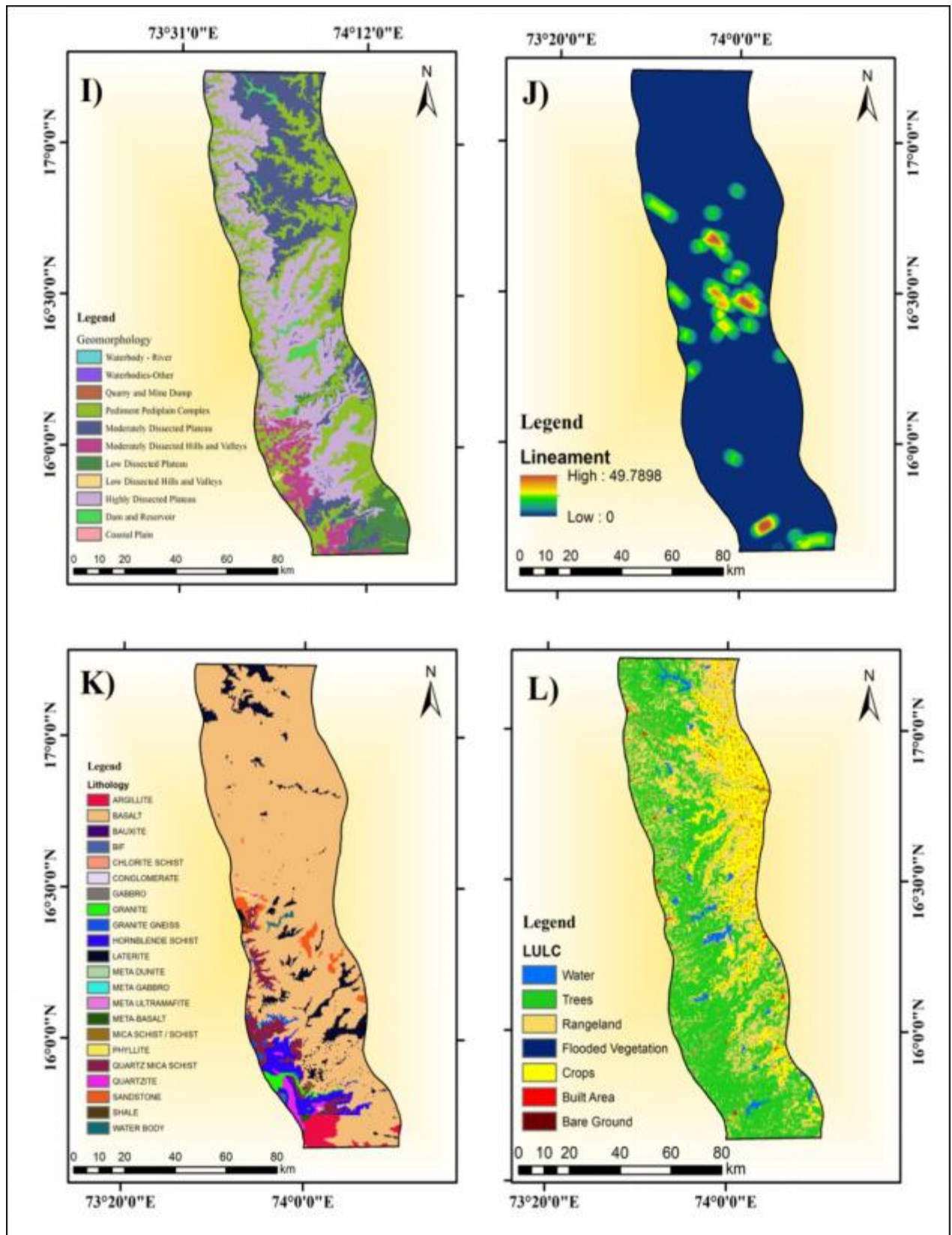


Fig. 2.3 I) Geomorphology , J) Lineament , K) Lithology , L) LULC

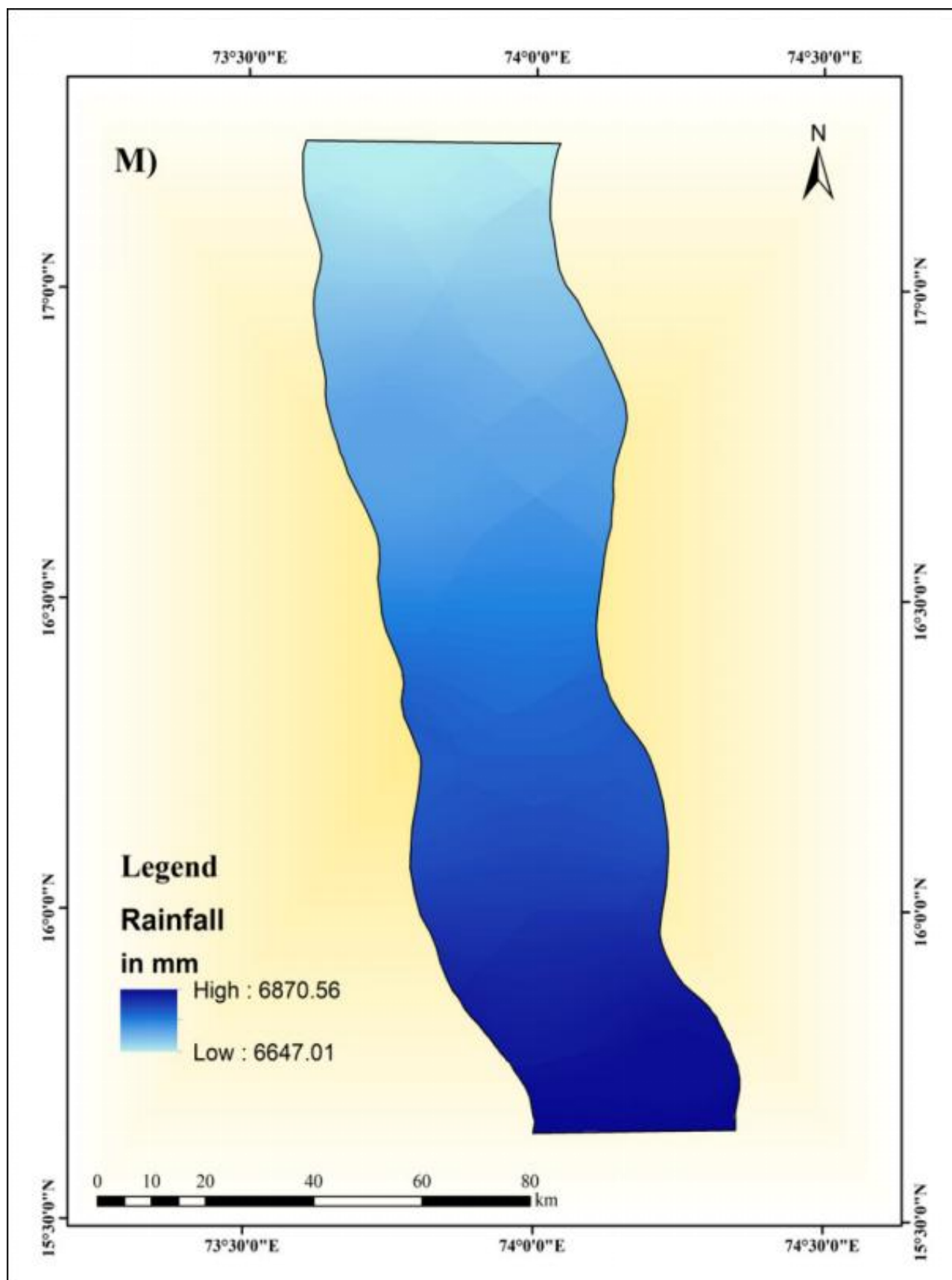


Fig 2.4 Rainfall in mm

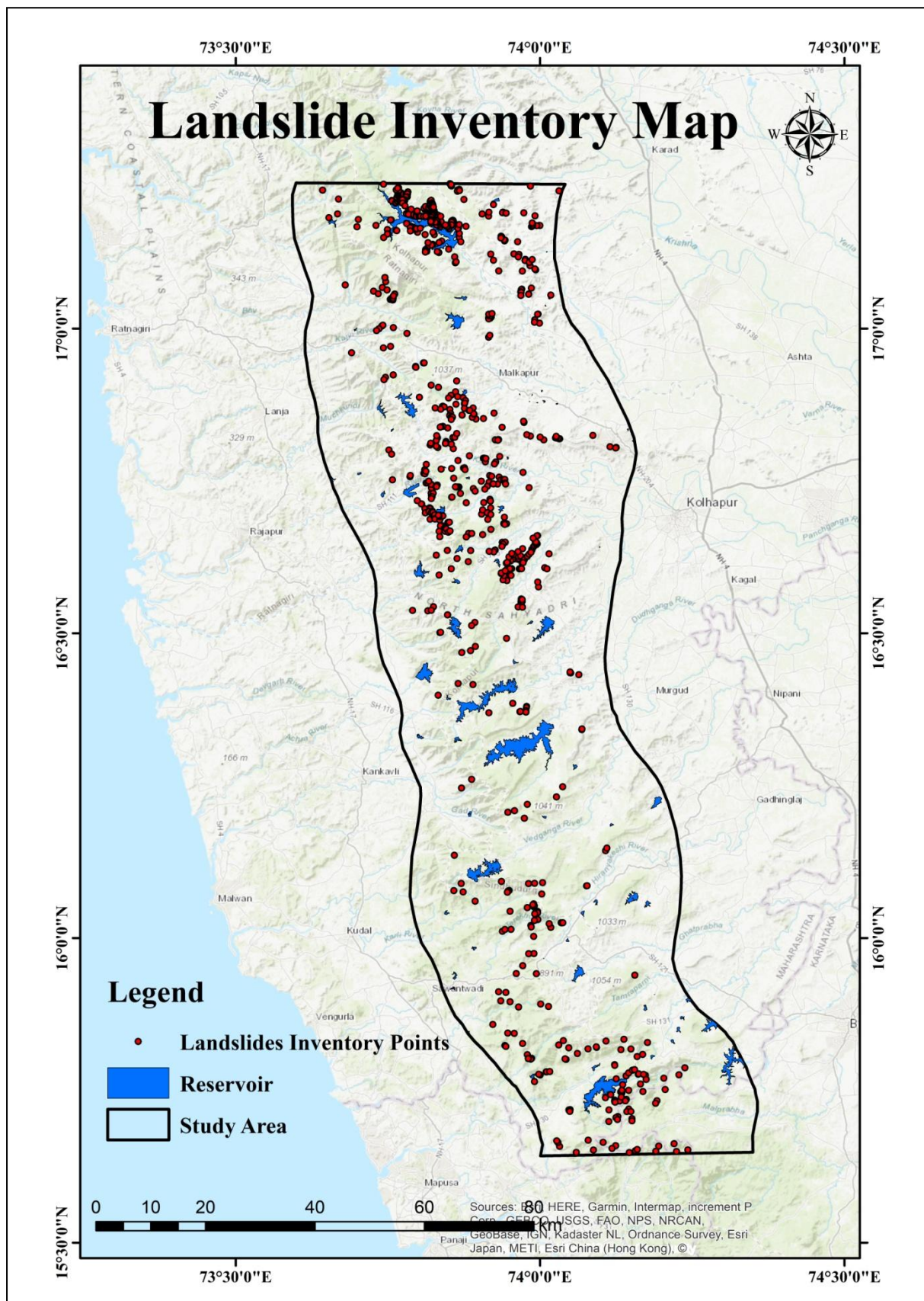


Fig 2.5 Landslide Inventory Map

Chapter 3: Methodology

3.1 Introduction

3.2 Methodological Framework

3.3 Data Collection and Integration

3.4 Landslide Inventory Data

3.5 Geospatial Data Acquisition

3.6 Factor Selection and Preprocessing

3.7 Data Splitting: Training and Testing Dataset

3.8 Landslide Susceptibility Modeling Using Machine Learning

3.9 Landslide Susceptibility Mapping

3.10 Summery

3.1 Introduction

Landslide susceptibility mapping (LSM) is a critical step in assessing landslide risk, which is essential for hazard mitigation and land-use planning in regions prone to landslides. Machine Learning (ML) algorithms, including Random Forest (RF), Support Vector Machine (SVM), and XGBoost, are employed to create an effective model for identifying and predicting landslide-prone areas. These models are particularly suited for handling complex datasets, such as those containing multiple geospatial layers with non-linear interactions.

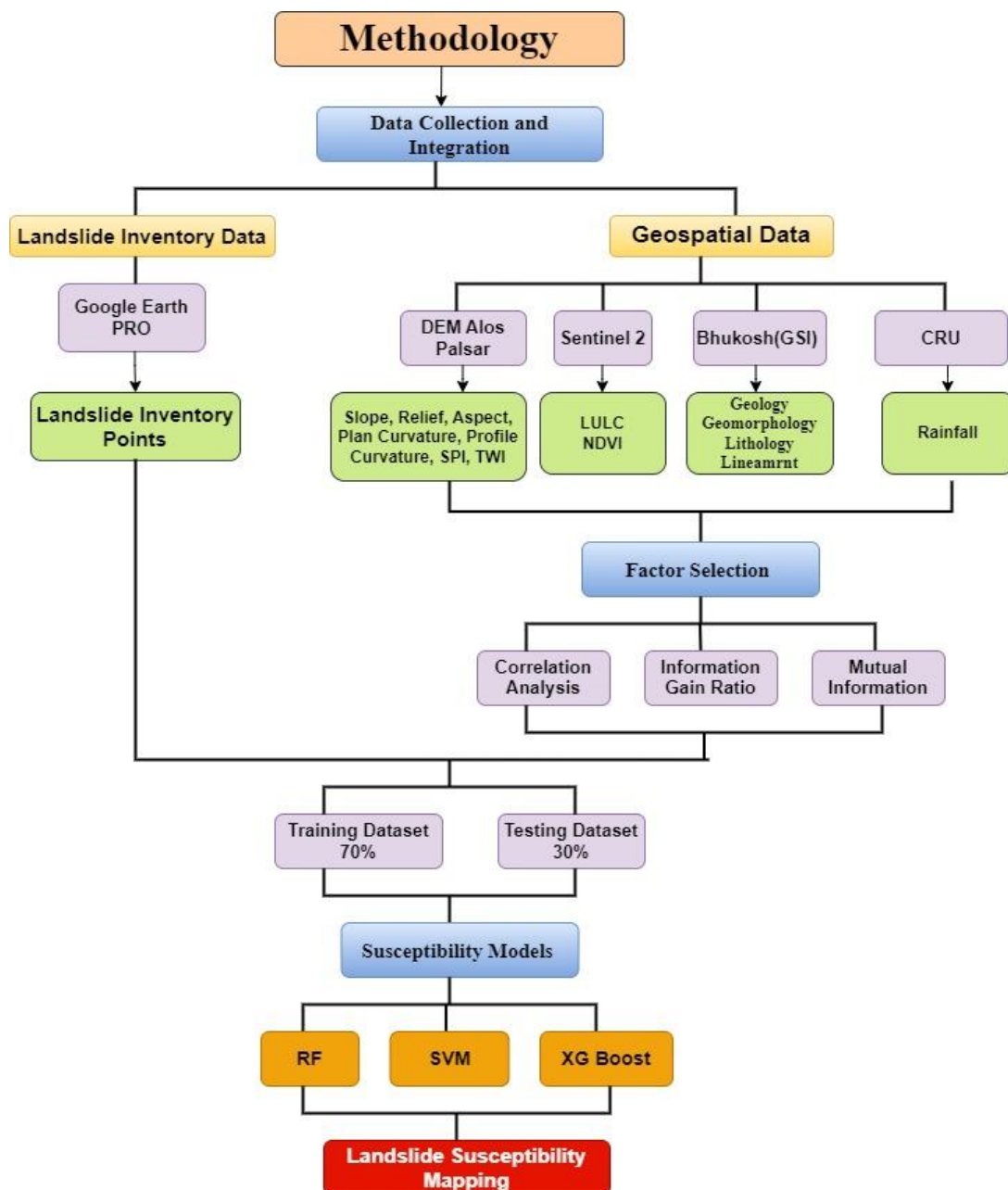


Fig 3.1 Methodology Chart

3.2 Methodological Framework

The methodology follows a structured approach consisting of the following key steps:

- **Data Collection and Integration**
- **Landslide Inventory Data**
- **Geospatial Data Acquisition**
- **Factor Selection and Preprocessing**
- **Training and Testing Data Splitting**
- **Susceptibility Modeling Using Machine Learning**
- **Landslide Susceptibility Mapping**

The detailed process is depicted in Figure 3.1 and explained below.

3.3 Data Collection and Integration

Data collection is the fundamental step in landslide susceptibility mapping. The study integrates two major datasets:

- **Landslide Inventory Data**
- **Geospatial Data**

These datasets are compiled from multiple sources, including Google Earth Pro, Remote Sensing Data, Geological Surveys, and Climate Data Archives.

3.4 Landslide Inventory Data

The landslide inventory serves as a crucial dataset for model training and validation. It consists of:

- **Historical Landslide Locations** (past events marked as landslide points)

- **Landslide-Free Locations** (randomly selected non-landslide points for training models)

Data Source: Google Earth Pro is used to extract landslide occurrence points based on visual interpretation and satellite imagery analysis.

3.5 Geospatial Data Acquisition

A variety of spatial datasets are collected from different sources to analyze landslide conditioning factors. The datasets include:

3.5.1 Digital Elevation Model (DEM) – ALOS PALSAR

- **Derived Features:** Slope, Relief, Aspect, Plan Curvature, Profile Curvature, Stream Power Index (SPI), and Topographic Wetness Index (TWI).
- **Purpose:** Used to assess the impact of terrain characteristics on landslide occurrences.

3.5.2 Sentinel-2 Satellite Data

- **Extracted Parameters:** Land Use Land Cover (LULC) and Normalized Difference Vegetation Index (NDVI).
- **Purpose:** NDVI helps in understanding vegetation patterns, which influence soil stability.

3.5.3 Geological Data (Bhuvan GIS - Geological Survey of India)

- **Geological Parameters:** Geology, Geomorphology, Lithology, and Lineament Density.
- **Purpose:** Helps in understanding rock strength and structural influences on landslides.

3.5.4 Climate Data (CRU - Climate Research Unit)

- **Parameter:** Rainfall Data
- **Purpose:** Rainfall is a key triggering factor for landslides, particularly in the Western Ghats region.

3.6 Factor Selection and Preprocessing

Once the raw geospatial data is collected, it undergoes preprocessing and factor selection to enhance prediction accuracy. Three techniques are used to select the most influential landslide factors:

- **Correlation Analysis:** Identifies relationships between variables to remove redundant data.
- **Information Gain Ratio:** Measures the importance of each factor in landslide susceptibility.
- **Mutual Information:** Determines the dependency between variables to improve feature selection.

The selected features are then standardized for better model performance.

3.7 Data Splitting: Training and Testing Dataset

The processed dataset is split into:

- **70% Training Data:** Used to train machine learning models.
- **30% Testing Data:** Used to evaluate model performance.

The split ensures balanced class distribution for better model generalization.

3.8 Landslide Susceptibility Modeling Using Machine Learning

Three distinct supervised machine learning models are employed in this study to generate the landslide susceptibility map for the Kolhapur district area. These models—Random Forest (RF), Support Vector Machine (SVM), and XGBoost—were chosen for their effectiveness in handling spatial data and their ability to model complex, non-linear relationships between input features and landslide occurrences. The following sections provide a detailed overview of each model:

3.8.1 Random Forest (RF)

- **Overview:** Random Forest is an ensemble learning algorithm that constructs multiple decision trees during training and merges their results to improve accuracy and prevent overfitting. This approach is particularly useful for handling large datasets with complex relationships among the features.
- **Mechanism:** It builds many individual decision trees using random subsets of the training data and random subsets of features. The final prediction is made by aggregating the predictions from each tree, typically through majority voting for classification tasks.
- **Advantages:**
 - Handles large datasets effectively and is less prone to overfitting compared to a single decision tree.
 - Provides feature importance measures, which can help in understanding the most influential variables in predicting landslides.
 - Can handle both categorical and numerical data with ease.

3.8.2 Support Vector Machine (SVM)

- **Overview:** Support Vector Machine (SVM) is a powerful classification algorithm used for both linear and non-linear classification tasks. It works by finding a hyperplane that best separates the different classes (in this case, landslide and non-landslide areas) in a high-dimensional space.
- **Mechanism:** SVM operates by mapping input features into a high-dimensional space and then constructing a hyperplane that maximally separates the two classes. For non-linear problems, a kernel trick, such as the Radial Basis Function (RBF) kernel, is employed to transform the input space and allow SVM to work efficiently in non-linear scenarios.
- **Advantages:**
 - Particularly effective in high-dimensional spaces, making it suitable for geospatial data where many factors influence landslides.

- Robust to overfitting, especially in high-dimensional data.
- Capable of handling complex, non-linear decision boundaries.

3.8.3 XGBoost

- **Overview:** XGBoost (Extreme Gradient Boosting) is a gradient boosting algorithm known for its high computational efficiency and predictive performance. It is an ensemble learning method that builds decision trees sequentially, where each subsequent tree attempts to correct the errors made by the previous ones.
- **Mechanism:** The algorithm minimizes a loss function by adjusting the weights of the individual trees using gradient descent, where each tree is trained to correct the mistakes of the previous tree. XGBoost has built-in regularization mechanisms that reduce the risk of overfitting and improve model generalization.
- **Advantages:**
 - Efficient in handling missing data and categorical features.
 - Provides superior handling of non-linear relationships and interactions between features.
 - Known for its speed and scalability, making it ideal for large datasets with many features.
 - Includes regularization to avoid overfitting, which is essential for ensuring robustness in landslide susceptibility models.

Each of these models has its strengths and weaknesses, and their performance is evaluated through cross-validation using a test dataset. The best-performing model is then selected for generating the final landslide susceptibility map. By utilizing these advanced machine learning techniques, this study aims to produce accurate and reliable predictions of landslide-prone areas, contributing to effective hazard management in the Kolhapur district.

3.9 Landslide Susceptibility Mapping

Once the models are trained and tested, the best-performing model is used to generate the Landslide Susceptibility Map. The map is categorized into five susceptibility zones:

- **Very Low Susceptibility**
- **Low Susceptibility**
- **Moderate Susceptibility**
- **High Susceptibility**
- **Very High Susceptibility**

GIS-Based Visualization: The final output is mapped using ArcGIS/QGIS to highlight high-risk areas.

3.10 Summery

This methodology provides a structured and data-driven approach for landslide susceptibility mapping using machine learning algorithms. By integrating geospatial data, historical landslide occurrences, and climate variables, this study ensures high predictive accuracy. The results will help in hazard mitigation and risk management for Kolhapur's hilly regions.

Chapter 4: Results and Discussion

4.1 Feature Selection

4.2 Model Evaluation Results

4.3 Comparison of Models

4.4 Feature Selection Impact

4.5 Discussion of Results

4.6 Summery

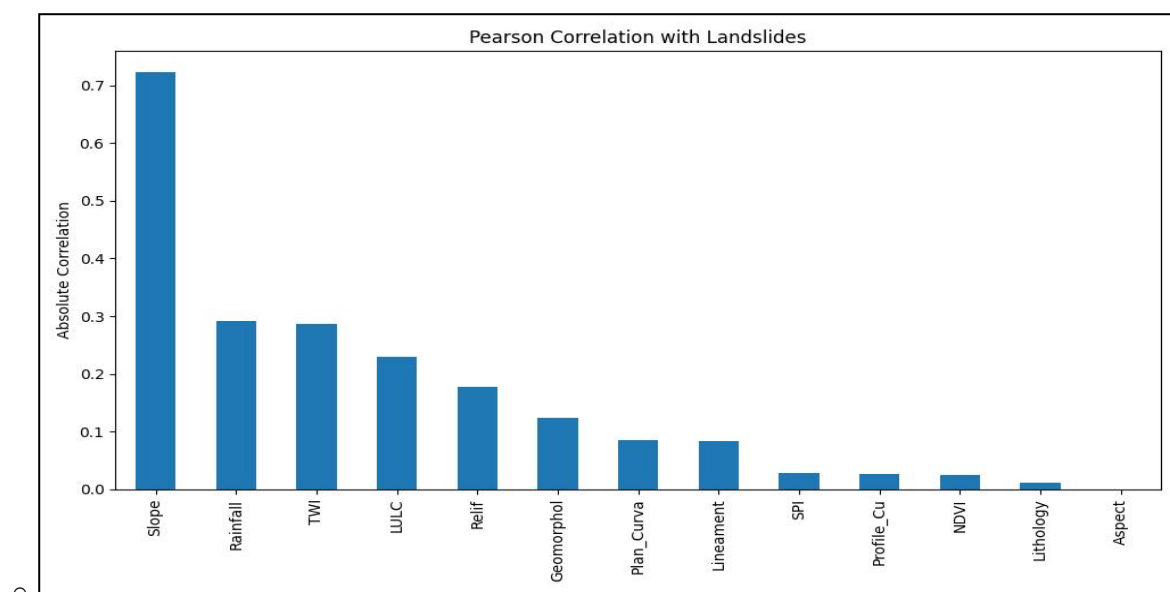
This chapter presents the results of the Landslide Susceptibility Mapping (LSM) project, where three machine learning models—Random Forest (RF), Support Vector Machine (SVM), and XGBoost—were applied to predict landslide susceptibility in the Kolhapur district. The results, including evaluation metrics and comparison of the models, are presented and discussed, followed by a detailed analysis of the feature selection process and its impact on model performance. Additionally, the results maps and feature selection graphs are included to visually support the findings.

4.1 Feature Selection

Effective feature selection is a crucial step in ensuring that the machine learning models capture the most relevant information from the data. The following feature selection techniques were used: Pearson Correlation, Mutual Information, and Information Gain Ratio.

- **Pearson Correlation:**

- The Pearson method identified the following features as most influential: Slope, Rainfall, TWI (Topographic Wetness Index), LULC (Land Use Land Cover), Relif (Relief), and Geomorphol (Geomorphology). These features are particularly important as they show direct correlations with landslide events. For instance, steep Slope values and intense Rainfall are known to trigger landslides.



- **Fig 4.1 Pearson Correlation**

- **Mutual Information:**

- The Mutual Information method revealed a broader selection of features: Slope, Rainfall, Geomorphol, Plan Curvature, Profile Curvature, LULC, Relif, Aspect, SPI (Stream Power Index), and TWI. Mutual Information uncovers non-linear dependencies between features and is effective for detecting intricate relationships in landslide occurrence.

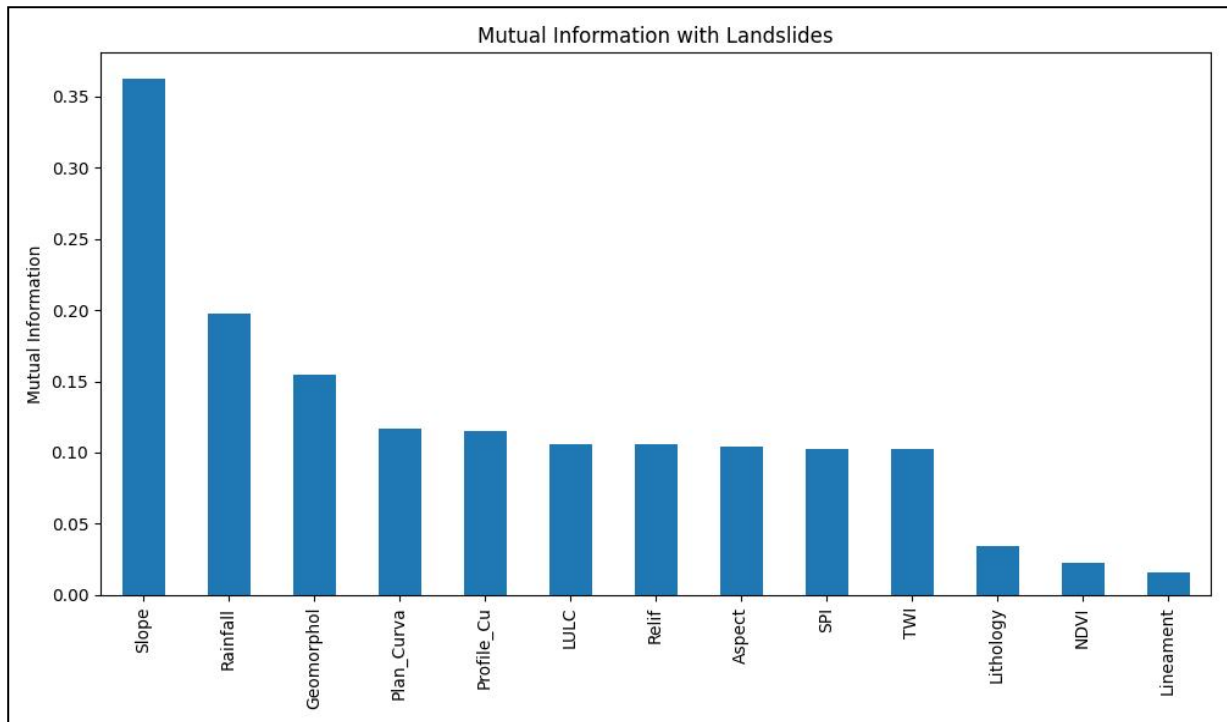
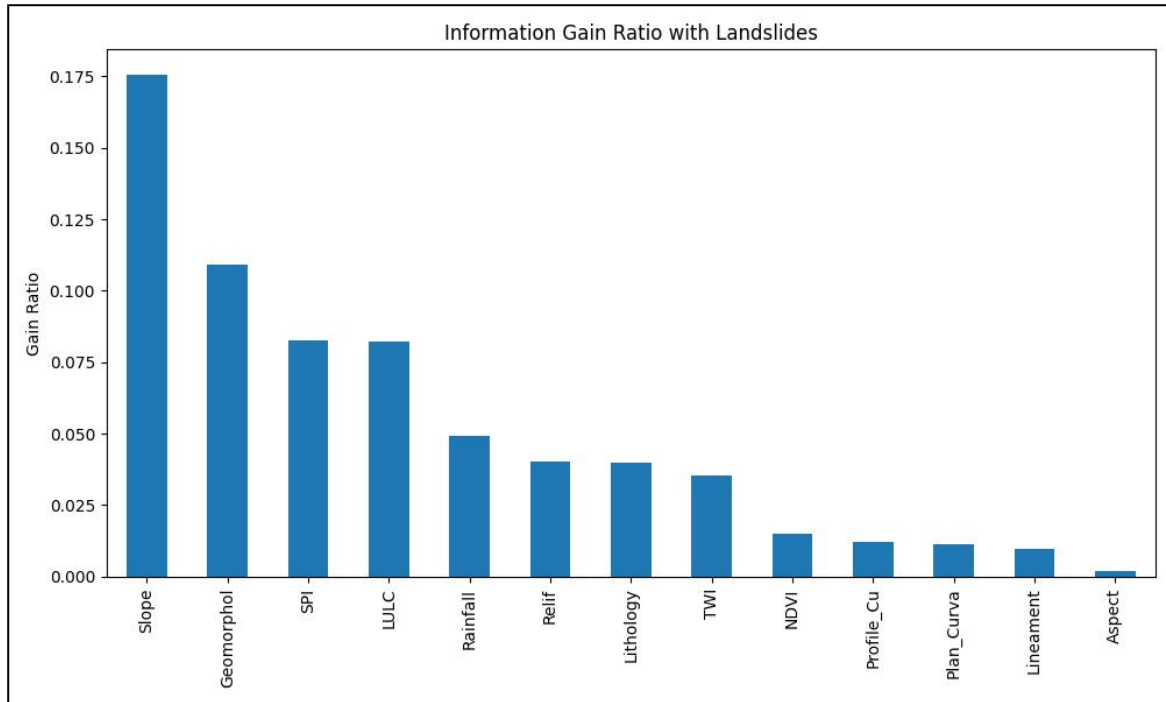


Fig 4.2 Mutual Information

- **Information Gain Ratio :**

- The Information Gain Ratio identified features such as Slope, Geomorphol, SPI, LULC, Rainfall, Relif, Lithology, TWI, NDVI (Normalized Difference Vegetation Index), and Profile Curvature as crucial. These features provide insights into both the static properties of the terrain and dynamic environmental factors.



• **Fig 4.3 Information Gain Ratio**

• **Final Selected Features:**

- After evaluating all three selection methods, the final set of features used in the modeling process included: Geomorphology, Plan Curvature, Lithology, Profile Curvature, Rainfall, LULC, Aspect, NDVI, Slope, SPI, TWI, and Relif. These features comprehensively represent the environmental, topographic, and vegetation factors that influence landslide susceptibility.

4.2 Model Evaluation Results

The three models—Random Forest (RF), Support Vector Machine (SVM), and XG Boost—were evaluated using multiple metrics, including ROC-AUC, Accuracy, F1-score, Kappa coefficient, and Confusion Matrix. The following results were obtained:

• **Random Forest (RF):**

- ROC-AUC: 0.9791 – This high value indicates that RF can efficiently distinguish between landslide and non-landslide areas.
- Accuracy: 92.81% – RF correctly predicted 92.81% of the landslide and non-landslide areas.

- F1-score: 0.9256 – The F1-score indicates a good balance between precision and recall.
- Kappa Coefficient: 0.8564 – The Kappa coefficient suggests strong agreement between the model's predictions and the actual observations.
- Confusion Matrix:
 - True Negatives (TN): 241
 - False Positives (FP): 28
 - False Negatives (FN): 8
 - True Positives (TP): 224

The confusion matrix indicates that the Random Forest model made relatively few misclassifications, with a low number of false positives and false negatives.

- **Support Vector Machine (SVM):**

- ROC-AUC: 0.9459 – The SVM model demonstrates good ability to separate landslide from non-landslide areas.
- Accuracy: 89.82% – While the accuracy is slightly lower than Random Forest, it still shows strong predictive performance.
- F1-score: 0.8966 – The F1-score reflects a good balance between precision and recall, though not as high as RF.
- Kappa Coefficient: 0.7970 – The Kappa coefficient shows a moderately strong agreement between the predicted and observed values.
- Confusion Matrix:
 - True Negatives (TN): 229
 - False Positives (FP): 40
 - False Negatives (FN): 11
 - True Positives (TP): 221

The confusion matrix indicates that SVM had a higher number of misclassifications, with 40 false positives and 11 false negatives. Despite this, the model still performed reliably.

- **XG Boost:**
 - ROC-AUC: 0.9812 – XG Boost achieved the highest ROC-AUC, indicating its superior performance in distinguishing between landslide and non-landslide areas.
 - Accuracy: 93.41% – The highest accuracy among all models, showing that XG Boost correctly predicted 93.41% of the cases.
 - F1-score: 0.9320 – The highest F1-score, indicating excellent precision and recall balance.
 - Kappa Coefficient: 0.8684 – The highest Kappa coefficient, suggesting a very high level of agreement between predictions and actual observations.
 - Confusion Matrix:
 - True Negatives (TN): 242
 - False Positives (FP): 27
 - False Negatives (FN): 6
 - True Positives (TP): 226

XG Boost achieved the lowest number of misclassifications with only 27 false positives and 6 false negatives, demonstrating its superior predictive power.

4.3 Comparison of Models

The performance of the three machine learning models—Random Forest, SVM, and XG Boost—was compared based on the evaluation metrics:

- ROC-AUC: XG Boost (0.9812) performed the best, closely followed by Random Forest (0.9791) and SVM (0.9459).

- Accuracy: XG Boost achieved the highest accuracy (93.41%), followed by Random Forest (92.81%) and SVM (89.82%).
- F1-score: XG Boost again had the highest F1-score (0.9320), indicating that it strikes the best balance between precision and recall, followed by Random Forest (0.9256) and SVM (0.8966).
- Kappa Coefficient: XG Boost had the highest Kappa coefficient (0.8684), showing a strong level of agreement, followed by Random Forest (0.8564) and SVM (0.7970).
- Confusion Matrix: XG Boost had the least misclassification with the lowest number of false positives and false negatives, followed by Random Forest and SVM.

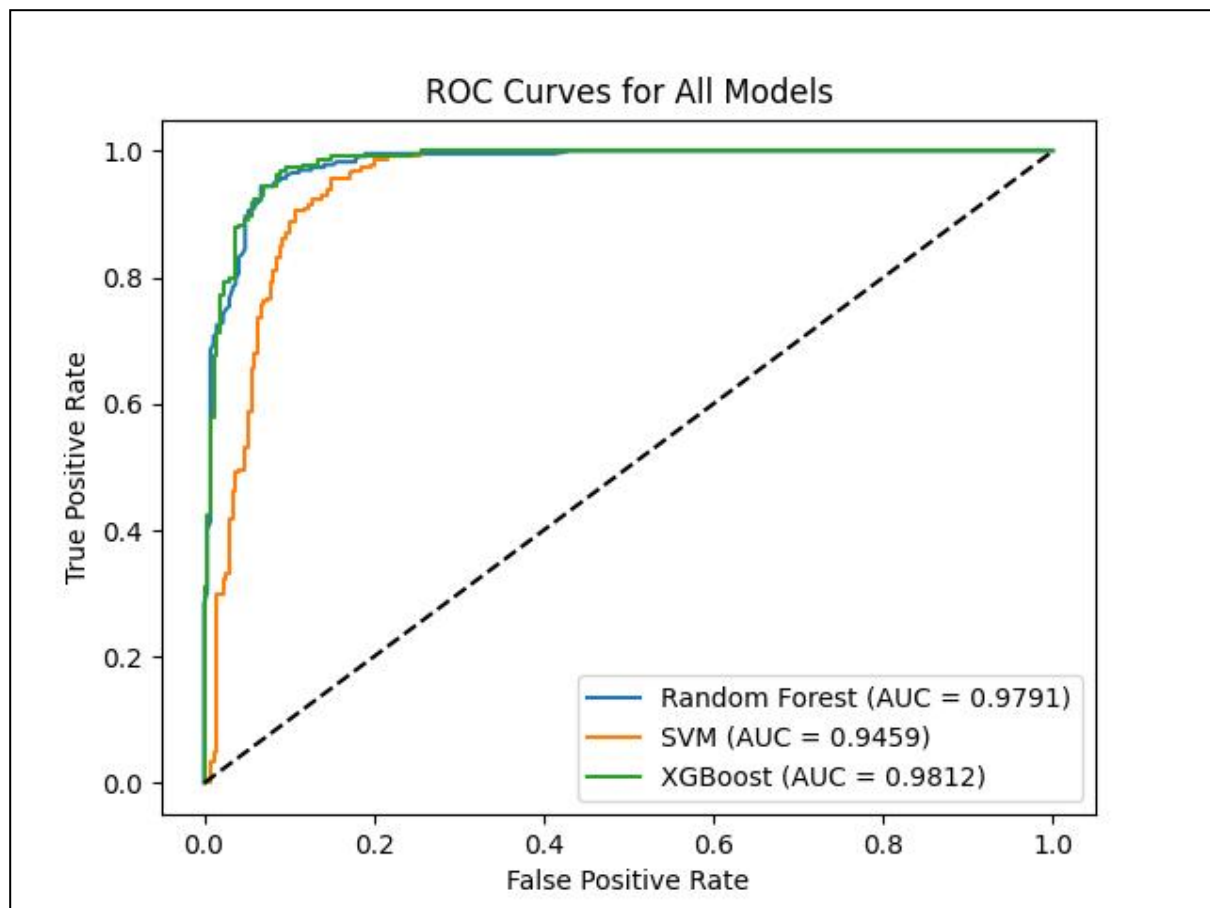


Fig 4.4 ROC Curves

These results confirm that XG Boost outperformed both Random Forest and SVM across all metrics, making it the most suitable model for landslide susceptibility mapping in the study area.

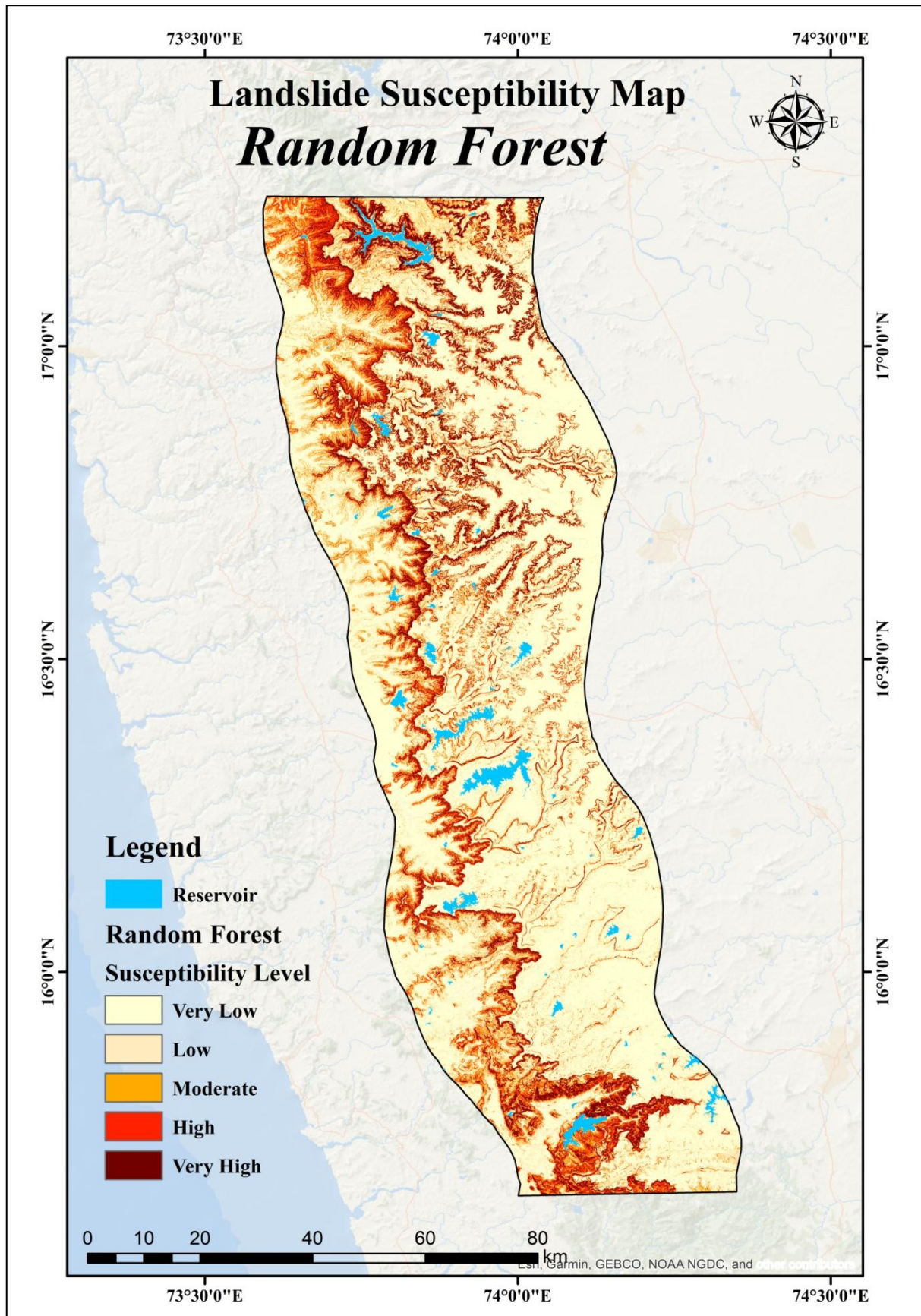


Fig 4.5 Random Forest Susceptibility Map

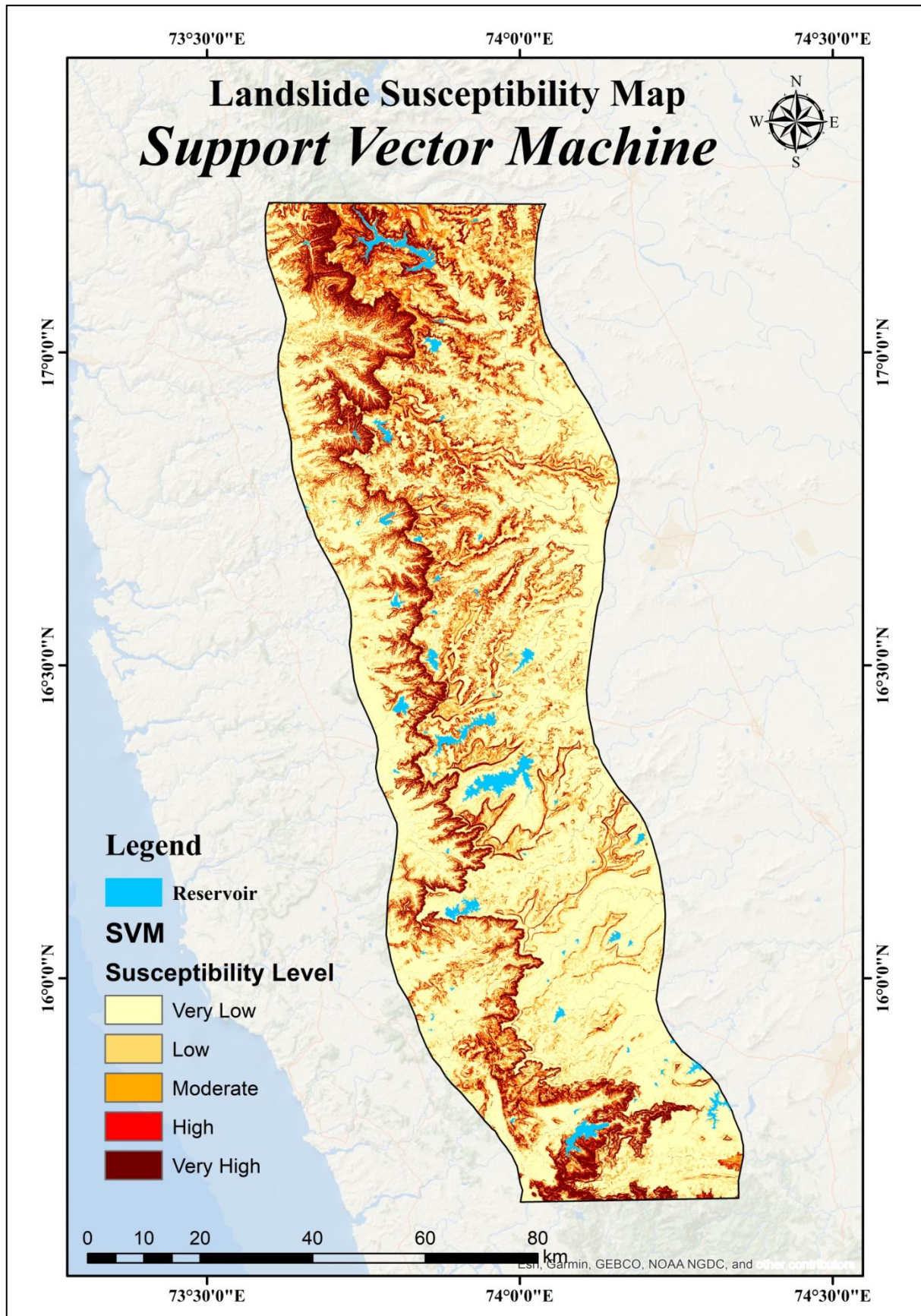


Fig 4.6 SVM Susceptibility Map

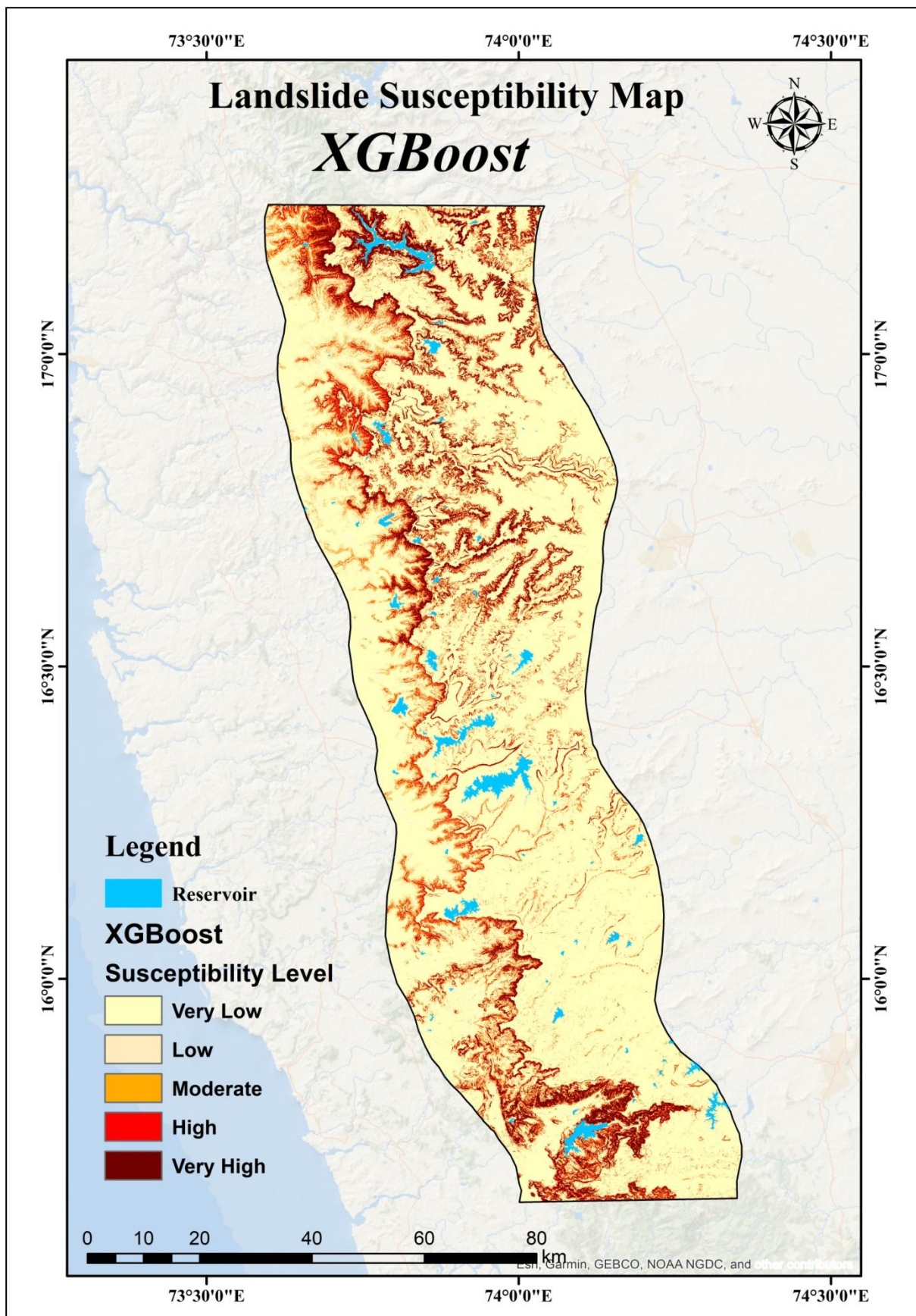


Fig 4.7 XG Boost Susceptibility Map

4.4 Feature Selection Impact

The inclusion of the selected features in the models significantly enhanced their ability to predict landslide susceptibility. For instance, Rainfall and Slope were found to be among the most important features, reflecting the influence of climatic and topographic factors on landslide occurrence. The final selected set of features, which included both physical factors (like Slope and Relief) and environmental factors (like LULC and NDVI), ensured that the models had access to all relevant data for accurate prediction.

Maps generated for each of the selected features visually represent their spatial distribution in the study area, highlighting regions most prone to landslides. These maps were created using GIS software (QGIS/ArcGIS), providing clear spatial patterns that correlate with the model predictions.

4.5 Discussion of Results

The results from the landslide susceptibility modeling in the Kolhapur district, conducted using **Random Forest (RF)**, **Support Vector Machine (SVM)**, and **XGBoost**, show the effectiveness of these machine learning models in predicting landslide-prone areas. A key component of this study involved the application of **hyperparameter tuning**, which significantly enhanced the models' performance. The hyperparameter optimization process allowed the models to perform at their best by selecting the optimal settings for each algorithm. This section discusses the impact of hyperparameter tuning on the models, compares their performance, and explores the implications of the results.

4.5.1 Impact of Hyperparameter Tuning on Model Performance

Hyperparameter tuning is a critical step in the machine learning workflow, as it enables models to better capture the underlying patterns in the data. In this study, grid search and cross-validation were used to tune the hyperparameters of each model, resulting in the selection of the best-performing parameters for each algorithm. The optimized models achieved higher predictive accuracy, improving the overall performance metrics.

Random Forest (RF):

The hyperparameter tuning process for **Random Forest** yielded the following best parameters:

- **n_estimators:** 300
- **max_depth:** None
- **min_samples_split:** 2
- **min_samples_leaf:** 1

These settings allowed the **Random Forest** model to generate a higher number of trees (300), improving its ability to model complex relationships. The lack of restriction on **max_depth** allows the trees to grow freely, which is beneficial when capturing detailed patterns in the data. The optimized **Random Forest** model achieved a **ROC-AUC** score of 0.9529, which reflects its robust performance in predicting landslide susceptibility. The accuracy, **F1-score**, and **Kappa coefficient** were also enhanced, highlighting the effectiveness of the model after hyperparameter optimization.

Support Vector Machine (SVM):

For **SVM**, hyperparameter tuning resulted in the following optimal parameters:

- **C:** 10
- **gamma:** 'scale'
- **kernel:** 'linear'

The **C** parameter controls the trade-off between achieving a low error rate and a high margin. A higher value of **C** (10) increases the model's flexibility in fitting the training data, which is beneficial for capturing the boundaries between landslide-prone and non-prone areas. The **gamma** value of 'scale' ensures that the kernel function is optimized for the data distribution. The **linear** kernel was chosen due to its simplicity and effectiveness in linearly separable datasets. The optimized **SVM** model achieved a **ROC-AUC** score of 0.9309, which, while lower than the other models, still indicates a good fit for the data.

XGBoost:

The **XGBoost** model, known for its high computational efficiency and ability to model complex relationships, was tuned with the following best parameters:

- **learning_rate**: 0.1
- **max_depth**: 3
- **n_estimators**: 100
- **subsample**: 1.0
- **colsample_bytree**: 1.0

These parameters allowed **XGBoost** to effectively balance model complexity and training efficiency. The **learning_rate** of 0.1 ensures that each step of the boosting process contributes meaningfully without overfitting the data. The **max_depth** of 3 indicates that each tree is relatively shallow, which helps prevent overfitting and allows the model to generalize well. The **ROC-AUC** score for the **XGBoost** model after hyperparameter tuning was 0.9621, the highest among all three models, highlighting its superior ability to predict landslide susceptibility.

4.5.2 Model Performance Comparison

After hyperparameter tuning, the three models were compared based on their **ROC-AUC**, **Accuracy**, **F1-score**, and **Kappa coefficient**. These metrics are essential for evaluating the models' ability to classify landslide-prone areas accurately and consistently.

- **XGBoost** showed the highest **ROC-AUC** score (0.9621), followed by **Random Forest** (0.9529) and **SVM** (0.9309). The higher **ROC-AUC** for **XGBoost** suggests that it was more capable of distinguishing between landslide-prone and non-prone areas, making it the most accurate model in terms of predictive power.
- In terms of **Accuracy**, **XGBoost** again led with 93.41%, followed by **Random Forest** (92.81%) and **SVM** (89.82%). These accuracy scores confirm that **XGBoost** had the highest proportion of correctly classified instances.

- The **F1-score** and **Kappa coefficient** also reflect **XGBoost**'s superior performance. **XGBoost** achieved an **F1-score** of 0.9320 and a **Kappa coefficient** of 0.8684, both higher than the other models. The **Kappa coefficient** indicates a strong agreement between the predicted and actual susceptibility classes, reinforcing the reliability of **XGBoost**'s predictions.
- **Random Forest**, while slightly behind **XGBoost**, still performed well with an **F1-score** of 0.9256 and a **Kappa coefficient** of 0.8564. These results demonstrate that **Random Forest** is a highly reliable model for landslide susceptibility prediction, especially when combined with hyperparameter tuning.
- **SVM**, while performing admirably, was outperformed by both **Random Forest** and **XGBoost**. With an **F1-score** of 0.8966 and a **Kappa coefficient** of 0.7970, **SVM** showed that it is still a viable model for landslide susceptibility but does not capture the complexity of the data as effectively as the ensemble methods.

4.5.3 Importance of Feature Selection and Hyperparameter Tuning

The combination of **feature selection** and **hyperparameter tuning** played a pivotal role in improving the performance of the machine learning models. The feature selection process identified key factors such as **Slope**, **Rainfall**, **TWI**, **LULC**, and **Geomorphology**, which were crucial for making accurate susceptibility predictions. These features helped the models capture the significant topographic, geological, and climatic factors influencing landslide occurrence.

Hyperparameter tuning further refined the models, enhancing their ability to predict landslide susceptibility in the Kolhapur district. By fine-tuning the model parameters, it was possible to optimize each algorithm's performance, ensuring that the models were not underfitting or overfitting the data. This step was particularly important for **XGBoost**, which benefited the most from hyperparameter optimization, leading to the highest overall performance.

4.5.4 Spatial Distribution of Landslide Susceptibility

The final landslide susceptibility maps generated by the models after hyperparameter tuning revealed clear delineations between **high-risk** and **low-risk** areas. **XGBoost**'s map, in

particular, exhibited the most detailed and accurate spatial distribution, with well-defined boundaries separating the regions that were most prone to landslides.

- **XGBoost** highlighted regions with steep slopes, high rainfall, and unstable geology as high-risk zones, while areas with lower slopes, better vegetation coverage, and stable geological features were classified as low-risk zones. These maps are valuable for planning and managing landslide risk in the study area.
- **Random Forest** provided a similar, though slightly less detailed, map, with well-defined high-risk zones, though some areas were more diffusely classified as moderate risk.
- **SVM** produced a map with broader zones of high and low susceptibility, reflecting its relatively less refined predictive power compared to the other models.

4.5.5 Implications for Landslide Risk Management

The results of this study have significant implications for landslide risk management in the Kolhapur district. The high-performance models, especially **XGBoost**, can help local authorities and planners identify vulnerable areas and take appropriate measures to mitigate the risk of landslides.

- **Infrastructure Development:** Areas classified as high-risk can be avoided for infrastructure development to minimize the risk of landslide damage to property.
- **Disaster Preparedness:** The susceptibility maps can serve as a foundation for early warning systems, helping authorities to issue timely alerts and evacuate residents from high-risk zones during extreme weather events.
- **Land Use Planning:** The maps can inform land use planning by identifying safe zones for agriculture, forestry, and other land uses, thereby minimizing the economic impacts of landslides.

4.5.6 Limitations and Future Work

While the models performed well after hyperparameter tuning, several limitations remain that could be addressed in future work:

- **Resolution of Input Data:** The spatial resolution of the input datasets (particularly the DEM) could be improved to provide more detailed susceptibility predictions, especially in areas with complex terrain.
- **Time-Series Data:** Incorporating time-series data on rainfall and vegetation changes could help capture the temporal dynamics of landslide susceptibility and improve model predictions.
- **Additional Feature Engineering:** Future studies could explore additional features, such as soil properties and historical landslide data, to enhance the models' accuracy further.

4.6 Summery

The combination of **feature selection** and **hyperparameter tuning** significantly improved the performance of **Random Forest**, **SVM**, and **XGBoost** for landslide susceptibility mapping. The optimized models, particularly **XGBoost**, provide valuable insights for landslide risk management in the Kolhapur district, offering a reliable tool for assessing and mitigating the impacts of landslides in this geologically sensitive region.

Chapter 5: Conclusion and Future Scope

5.1 Conclusion

5.2 Future Scope

5.1 Conclusion

The study on Landslide Susceptibility Mapping (LSM) using machine learning (ML) techniques in the Kolhapur district has successfully achieved the objectives set forth for the project. The primary goal of the research was to assess the landslide susceptibility in a geologically complex and highly rainfall-prone region by leveraging advanced ML models—Random Forest (RF), Support Vector Machine (SVM), and XGBoost—along with effective feature selection and hyperparameter tuning. The methodology adopted for this study has yielded insightful results that contribute to better understanding landslide susceptibility and offer valuable tools for decision-makers involved in disaster risk management.

Key findings of the study include:

- **Feature Selection:** A variety of factors, including Slope, Rainfall, Topographic Wetness Index (TWI), Land Use Land Cover (LULC), Geomorphology, and Vegetation Index (NDVI), were identified as significant predictors of landslide susceptibility in the study area. Through multiple feature selection techniques, including Pearson correlation, Mutual Information, and Information Gain Ratio, the most influential features were selected, providing a strong foundation for the ML models.
- **Hyperparameter Tuning:** The application of hyperparameter optimization using grid search and cross-validation improved the performance of all three machine learning models, resulting in the identification of the best parameters for each model. This process was essential for enhancing the predictive power and generalization ability of the models.
- **Model Performance:** The evaluation of the models' performance through ROC-AUC, Accuracy, F1-score, and Kappa coefficient demonstrated that the XGBoost model outperformed Random Forest and SVM in terms of predictive accuracy, achieving the highest ROC-AUC (0.9621) and Accuracy (93.41%). Despite its lower performance, SVM remained a viable model, providing a more computationally efficient alternative for landslide susceptibility prediction.
- **Susceptibility Mapping:** The final susceptibility maps generated by the models highlighted high-risk areas for landslides, with well-defined zones for very high, high,

moderate, low, and very low susceptibility. These maps serve as valuable tools for land-use planning, disaster risk management, and infrastructure development in the region.

- **Impact on Risk Management:** The findings have important implications for landslide risk management in the Kolhapur district. By identifying vulnerable areas, the susceptibility maps can guide authorities in disaster preparedness, early warning systems, and infrastructure planning. This study also emphasizes the importance of integrating remote sensing data and machine learning algorithms for large-scale environmental monitoring and risk management.

In conclusion, this research provides a comprehensive and data-driven approach to landslide susceptibility mapping, utilizing machine learning algorithms to achieve high accuracy in identifying landslide-prone areas. The results have the potential to support decision-making processes related to landslide risk mitigation, urban planning, and disaster management in the study area.

5.2 Future Scope

While this study has provided valuable insights into landslide susceptibility mapping using machine learning, several aspects of the research can be expanded upon in future work. The following are potential areas for further improvement and exploration:

1. Improved Data Resolution:

- The study relied on datasets with moderate resolution, such as the ALOS PALSAR DEM (12.5 m resolution) and Sentinel-2 imagery (10 m resolution). Future studies could utilize higher-resolution datasets, such as LiDAR data or WorldDEM, to capture finer terrain details, which would improve the precision of the landslide susceptibility maps, especially in regions with highly complex topography.

2. Incorporation of Temporal Data:

- Landslide susceptibility is dynamic and can change over time due to factors such as vegetation growth, land use changes, and climate variations. The inclusion of time-series data (e.g., seasonal rainfall data, vegetation changes,

or historical landslide data) would allow for a better understanding of how landslide susceptibility evolves over time and improve the prediction models' temporal accuracy.

3. Additional Features for Model Enhancement:

- Future research can explore additional geological features, such as soil properties, rock strength, and fault lines, which may play an important role in landslide susceptibility. Including these factors in the modeling process could enhance the robustness and reliability of the predictions.
- Soil moisture and seismic data could also be incorporated as additional features, as these factors can influence landslide occurrence during heavy rainfall or seismic events.

4. Hybrid Modeling Approaches:

- A potential avenue for further research is the development of hybrid models, which combine the strengths of multiple machine learning techniques to improve predictive performance. For example, combining XGBoost with Random Forest or integrating Deep Learning algorithms such as Convolutional Neural Networks (CNNs) could further enhance the accuracy of landslide susceptibility mapping, particularly in areas with highly non-linear relationships between features.

5. Integration with Other Hazard Models:

- In addition to landslides, the study could be extended to integrate other natural hazards, such as floods or earthquakes, to create a more comprehensive multi-hazard risk assessment. Such integration would provide decision-makers with more holistic information for planning and risk management.
- Risk-weighted maps that combine landslide susceptibility with flood or seismic hazard maps would be particularly valuable for regions that face multiple hazards, allowing for more effective mitigation strategies.

6. Real-Time Monitoring and Early Warning Systems:

- The integration of real-time satellite data and IoT-based ground sensor networks could allow for continuous monitoring of landslide-prone areas and provide more accurate early warning systems. Machine learning models can be updated regularly with new data to improve prediction accuracy and enable more timely warnings to mitigate the impact of landslides.

7. Expanding the Geographic Scope:

- While this study focused on a specific region in Kolhapur, extending the methodology to other regions with different geological and climatic conditions would test the generalizability of the models. A broader geographic scope could offer more insights into the applicability of the models to various environments and further validate their robustness in different contexts.

8. Stakeholder Engagement and Practical Implementation:

- Future work should focus on involving local authorities, policymakers, and communities in the landslide risk assessment process. The models and maps produced should be made accessible to stakeholders in user-friendly formats, such as interactive web-based platforms or mobile applications, to ensure that the results are effectively utilized for real-world decision-making.
- Collaboration with disaster management agencies could lead to the development of interactive GIS-based tools for real-time landslide monitoring and early warning, integrating various data sources (e.g., weather forecasts, seismic activity, and terrain conditions).

9. Use of Advanced Artificial Intelligence:

- The use of deep learning and neural networks in landslide susceptibility mapping could further improve model performance, particularly for detecting complex patterns in large datasets. Advanced AI techniques, such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) models, could be explored to predict landslide occurrences based on sequential data (e.g., rainfall intensity over time).

5.3 Final Remarks

This study highlights the effectiveness of machine learning models, particularly XGBoost, in landslide susceptibility mapping for the Kolhapur district, providing crucial insights for landslide risk assessment and management. The integration of hyperparameter tuning and feature selection enhanced the accuracy and robustness of the models, making them suitable for real-world applications in disaster management and urban planning.

However, there is still considerable potential for improvement and expansion. Incorporating higher-resolution data, temporal variables, additional features, and hybrid modeling approaches could further improve the prediction accuracy. Additionally, integrating these models into practical risk management tools and early warning systems would significantly contribute to mitigating the devastating effects of landslides in vulnerable regions.

In summary, this research sets the stage for future advancements in landslide susceptibility modeling, offering a foundation for more accurate, dynamic, and actionable predictions that can inform better decision-making and reduce the risk of landslides in hilly and geologically sensitive areas.

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